

PYTHON MINI PROJECT

Smart Expense Tracker

A beginner-friendly web application to record, manage, and analyze daily expenses

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Why Track Your Expenses?



The Problem

- Manual tracking is error-prone and time-consuming
- No visibility into spending patterns
- Paper records are easily lost

The Solution

- Digital, secure expense storage
- Instant search and category-wise analysis
- Visual reports for smarter decisions

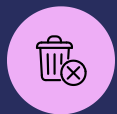
What This Project Does



Add Expense



View Expenses



Delete Expense



Search Expense



Category Report



Pie Chart

Tools & Technologies Used



Frontend

- HTML5
- CSS3
- Bootstrap



Backend

- Python 3
- Flask Framework



Database

- SQLite
- Lightweight, file-based



Visualization

- Matplotlib
- Pie chart generation



How the Application Works



Open App



Add/View Expenses



Save to SQLite



Generate Reports

The application follows a simple MVC pattern — user input is processed by Flask, stored in SQLite, and visualized using Matplotlib.

Database Schema

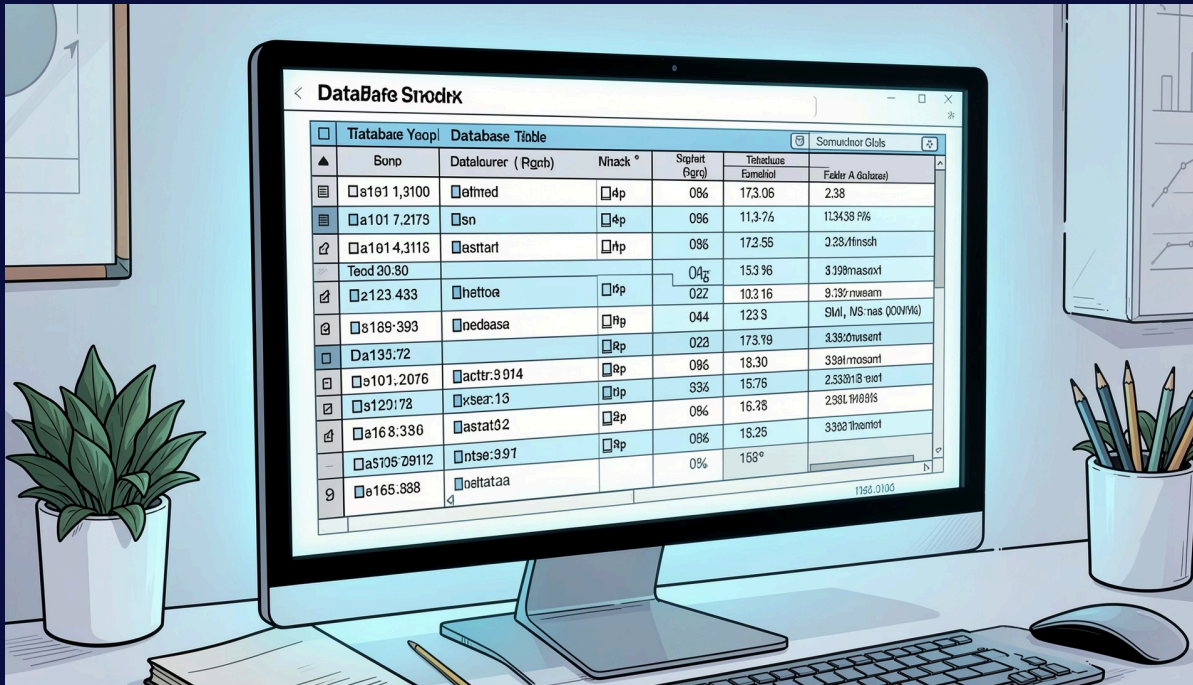


Table: expenses

Field	Type	Constraint
id	Integer	Primary Key
amount	Real	—
category	Text	—
description	Text	—
date	Text	—

The screenshot displays a dashboard for 'Apity' with a sidebar menu on the left containing: Dashboard, Projects, Accounts, Expenses, Projects Map, and Settings. The main content area is titled 'Expense' and includes a 'Expense Summary' section with the following data:

- Expense Limit: \$9,360,000/May
- Expense: \$2,820
- Expense Summary: 26% (with a green 'OK' indicator)
- Expense Limit: \$7,720,400,000 (with a green 'OK' indicator)

Below the summary, there are three charts:

- Expense Breakdown by Department**: A bar chart showing expenses for various departments (Sales, Marketing, HR, Finance, IT, R&D, Legal, Compliance, Other) with a total of \$2,820,000.
- Expense Breakdown by Category**: A bar chart showing expenses for categories (Travel, Accommodation, Food & Beverage, Entertainment, Transportation, Other) with a total of \$2,820,000.
- Expense Breakdown by Month**: A line chart showing expenses from January to May, with a total of \$2,820,000.

[illegible]

EXPENSE CATEGORIES

Category	Percentage
DIRECT INCOME	31.1%
PASSED INTERESTS	51.5%
COURT	30% 30%
CASH	10.5%
POLARIS	5.5%
TON	2.5%
COLLEGE	2.5%
ESTATE	2.5%
OFF-EXPENSE EARNINGS	7%

Pie Chart Report



Summary & What's Next

Conclusion

A Flask web app for managing daily expenses with SQLite storage and Matplotlib charts.

Future Scope

- User login and authentication
- Budget alerts and monthly reports
- Export to PDF / CSV
- Mobile application
- Cloud database integration

Thank You